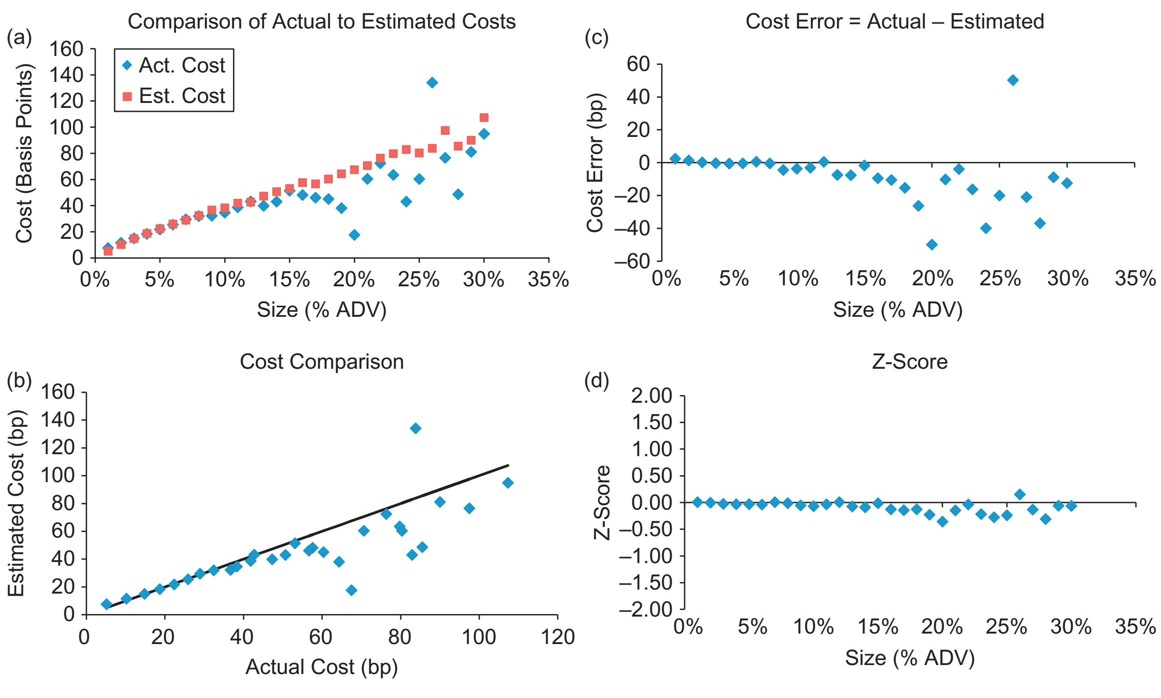
Our error analysis consists of estimating the market impact cost for each of the remaining data points. We then grouped these data points into 1% ADV categories and graphically compared the results (Figure 5.8a). We observed a very strong and accurate fit for order size up to about 15% ADV and for sizes of 20% to 30% it appears that we are overestimating the actual cost. Figure 5.8b is an xy-scatter plot of estimated cost (y-axis) as a function of actual cost (x-axis). This graph shows that our market impact model is accurate for costs up to approximately 40 to 50 bp. Again, the model seems to overestimate costs for larger more expensive orders. Figure 5.8c plots the actual error measured as the difference between estimated cost and actual cost by order size. This figure gives the appearance that the model begins to overestimate costs at around 10%−15% ADV.

The difference emerging here between estimated cost and actual cost is not concerning for the larger more expensive orders due to survivorship bias and opportunistic trading. That is, investors are more likely to complete the larger size orders in times of favorable price momentum and a lower cost environment. Furthermore, investors are more likely to increase the original order in times of more favorable prices. In times of adverse price movement and a higher trading cost environment investors



■ **Figure 5.8** (a) Comparison of Actual to Estimated Costs; (b) Cost Comparison; (c) Error Analysis; (d) Z-Score Analysis.